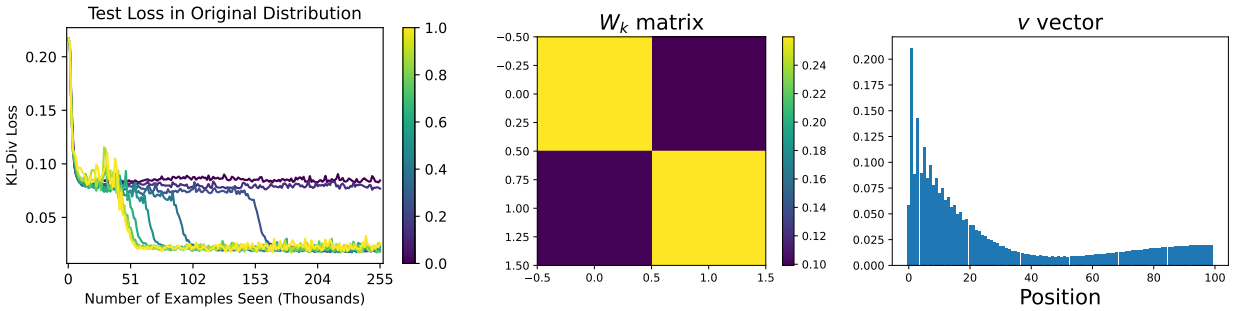

(a) (b)

Figure 4: (**a**) Transformers trained on different data distributions and evaluated at the original distribution. Color displays a smooth interpolation between data distributions of uninformative unigrams strategy (purple, 0) and Bayes optimal unigrams (yellow, 1). When there is not much signal from unigrams, learning progresses faster without long plateaus. See in Appendix B for a description of the data distributions. (**b**) Training of the minimal model in In-Context Learning Markov Chains with $k = 2$ states. (**left**) The heatmap of the 2nd layer (W_k matrix) that learns to be close to diagonal. (**right**) The values of the positional embeddings (1st layer) that display the curious even/odd pattern. Timestep corresponds to a phase when the model has started implementing the bigrams solution, but has not converged yet.

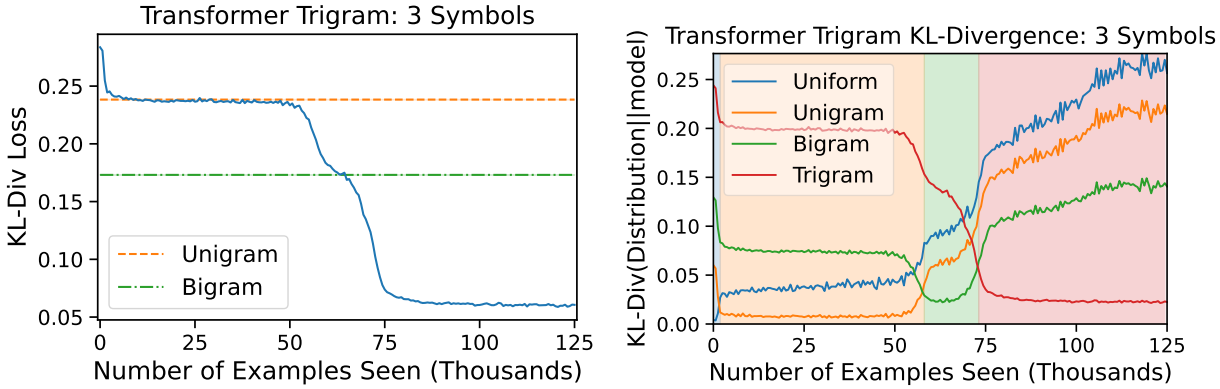

Figure 5: Three-headed transformer trained on In-Context Learning 3-grams (trigrams), with context length 200. **Left:** Loss during training. The model hierarchically converges close to the Bayes optimal solution. **Right:** KL divergence between the model and different strategies during training. As we observe, there are 4 stages of learning, each of them corresponding to a different algorithm implemented by the model.

dominating attention (that is, if W_Q is unimportant in that layer), we can define an analogue of the minimal model: $\hat{v}_i = \text{softmax}(e_{x_i} W_k v^T)$. Unlike v in the minimal model, $\hat{v}_i$ depends on the token at position i, but in practice, during most of training, $\hat{v}_i$ is similar regardless of the value of x_i. As can be seen on the top of Figure 3 (see also Figure 12), shortly before the second drop in the loss as the induction head forms around $t = 92$, the positional embeddings start showing the same even/odd pattern for the first positions.

If unigram solution is not correlated, it is skipped. By tweaking the data distribution, we can probe different parts of learning. In particular, when we focus on a special case of distribution where there is no unigrams solution, transformers learn much faster and do not plateau for long in the intermediate stage - see Figure 4a. The calculations also seem to confirm this (Corollary A.2 in the Appendix): the gradient of v at the first step evaluated on this special distribution shows that, in contrast to the default case, there is

no "heavy" preference towards the first coordinate. Then, once there is enough signal from the second layer, training progresses much more smoothly.

We now show that two-stage learning is not only necessary in some sense, but also sufficient to reach the bigrams solution. We analyze the optimization process of the minimal model and discover, that much like in the experiments with the transformers, there are separate phases that correspond to learning of different parts of the model. In particular, our analysis is in a simplified setting where the data distribution changes in the second step; the transition matrices are no longer sampled uniformly, but they are instead the "hard" examples of this task and contain the most "information" about the solution. This idea is closely related to the technique of curriculum learning (Bengio et al., 2009) and is naturally motivated in many settings. In this setting, we are able to show that two steps of gradient descent will reach the bigrams solution.

Proposition 3.2. *(Informal) Consider the minimal model of eq.* (8) *being trained with online SGD on the margin loss* (7) *with in-context loss* (6)*. With appropriate choice of learning rates and weight decay in the 2nd layer, two steps of gradient descent with curriculum learning reach the bigrams solution.*

The idea of the proof is that a first step of gradient descent with a small learning rate can align the 2nd layer, while a second step on "hard" examples can learn to identify perfectly the relevant relative embedding. The formal statement and its proof can be found in Appendix A. An interesting by-product of the analysis is that the learning rate at the first step needs to be an order of magnitude larger than at the second step, perhaps explaining why the second phase of learning in the experiments takes significantly more time. It is worth noting that, while this is a simplified setting, it goes beyond NTK-based (Jacot et al., 2018) analyses where the representations do not change much and it crucially involves more than one step which has been a standard tool in the analysis of feature learning in deep learning (Ba et al., 2022).

3.3 Beyond Bigrams: n-gram Statistics

We also investigated the performance of transformers on learning in-context n-grams for $n > 2$; in particular, 3-grams. We trained attention-only transformers with three heads in each layer by minimizing the in-context cross entropy loss with the Adam optimizer. As can be seen in Figure 5 (left), the model eventually converges to the Bayes optimal solution. Interestingly, as in the case of Markov Chains, the model displays a "hierarchical learning" behavior characterized by long plateaus and sudden drops. In this setup, the different strategies correspond to unigrams, bigrams and trigrams, respectively. This is presented clearly on the right of Figure 5, where we plot the similarity of the model with the different strategies and it exhibits the same clear pattern as in the case of $n = 2$. Single attention headed models could not achieve better performance than bigrams. We leave a more thorough investigation for future work.

4 Conclusion and Discussion

In this work, we have introduced a simple learning problem which serves as a controlled setting for understanding in-context learning and the emergence of (statistical) induction heads. Induction heads are a common motif in LLMs, which suggests the natural direction of exploring to what extent our various findings extend to natural language training data. Notably, on the way to solving the ICL-MC task, networks pass through a sequence of well-characterized discrete incomplete solutions. Simple but incomplete solutions may be commonplace in language modeling and other rich learning settings; for any such solution, one can ask to what extent its presence speeds up or slows down the formation of more complex circuits with higher accuracy.

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